Strategic Fee Choice and Transaction-Priority Allocation under Incomplete Information

A Three-Round Dynamic Bayesian Game with Private Urgency and Waiting

Lin Zhang · Zijie Wang | Team 7

Course instructor: Professor Luyao Zhang

昆山杜克大学 DUKE KUNSHAN UNIVERSITY

A nonbinding coordination signal improves transaction-priority allocation, but this comes with higher fee expenditure and lower total user welfare in our baseline simulation.

33.36% → 5.20%
Same-fee collision

87.11% → 97.99%
Priority alignment

01 Question & model

Research question

How do private urgency and accumulated waiting affect transaction-priority allocation, and can a nonbinding coordination signal improve it?

Setting & Gap

Two users with private Urgent (U) or Patient (P) types choose Low (L), Medium (M), or High (H) fees across three connected rounds. Higher fees receive confirmation priority; equal fees use a random tie-break. Unconfirmed users accumulate waiting, changing later incentives. Existing work provides foundations for Bayesian games and transaction-fee mechanism design. We connect private urgency, dynamic waiting, discrete fee choice, and coordination within one allocation problem.

Strategic model

Players: two users
Private type: $\theta_i \in \{U, P\}$
Rounds: $t = 1,2,3$
Action: $\alpha_{i,t} \in \{L, M, H\}$

Allocation Rule

$Higher\ fee \rightarrow confirmed$
$Same\ fee \rightarrow 50-50\ tie$
$Confirmed$: $w_{i,t+1} = 0$
$Unconfirmed$: $w_{i,t+1} = w_{i,t} + 1$

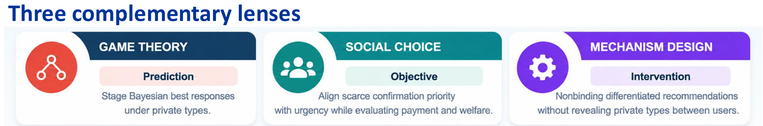
Baseline Calibration

$c(L) = 1, c(M) = 2, c(H) = 3$
$(V_U, d_U) = (8,2), \quad (V_P, d_P) = (5,1), \quad \Pr(U) = 0.6$

Solution Concepts

Full model: finite-horizon dynamic Bayesian game
Strategic benchmark: stage Bayesian best response
Against H, H is a strict best response if $V_\theta - c(H) + d_\theta(w+1) > 0$.
Under our calibration, this holds for both types:
(H, H) is a stage Bayesian Nash equilibrium.
Strategic tension: individually rational fee escalation can still produce a same-fee priority tie.

Three complementary lenses



Allocation comparison

Scarce resource: Transaction confirmation priority
Private information: Urgent/Patient type
Actions: Low / Medium / High fee
Comparison: Same allocation rule under three information and coordination conditions

02 Design & evidence

From Private Types to Observable Outcomes

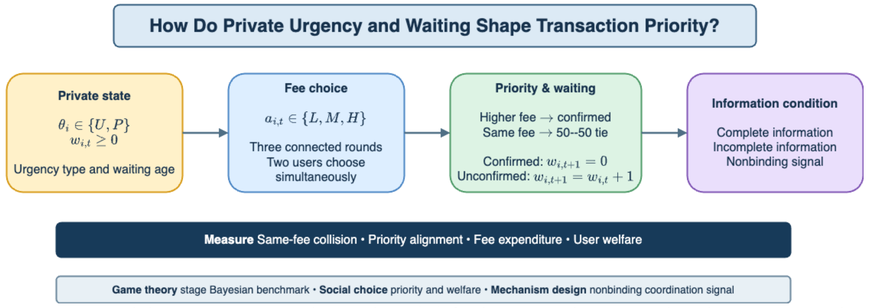


Figure 1. Research design. Private urgency and waiting shape L/M/H fee choices across three connected rounds. We compare complete information, incomplete information, and a nonbinding coordination signal.

Computational Comparison

We simulate **100,000 three-round sessions** under three conditions: **Complete information**, **Incomplete information**, and **Incomplete information + Signal.** The simulation policy responds to urgency and accumulated waiting and is evaluated separately from the stage equilibrium prediction.

The Signal Improves Priority Allocation

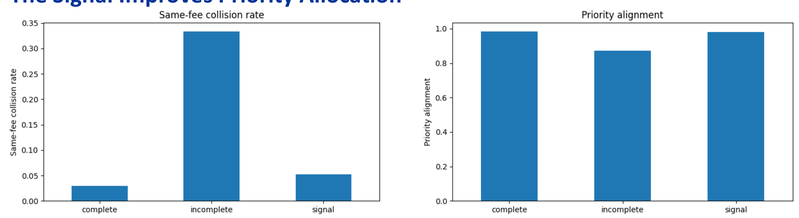


Figure 2. Allocation outcomes. Relative to incomplete information, the signal reduces same-fee collision from 33.36% to 5.20% and raises priority alignment from 87.11% to 97.99%.

Better Coordination Comes with a Cost

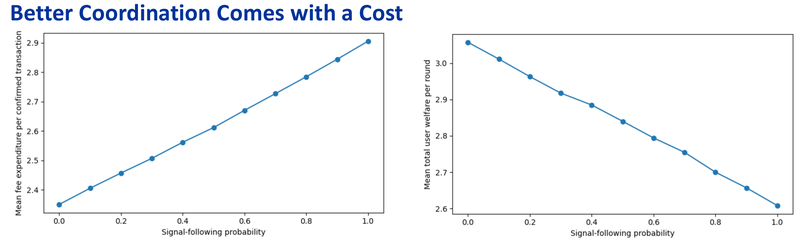


Figure 3. Signal-compliance sensitivity. As signal-following probability increases, mean fee expenditure rises while mean total user welfare falls under the baseline calibration.

At baseline compliance ($\rho = 0.85$), the signal increases mean fee expenditure from 2.352 to 2.812 and reduces mean total user welfare from 3.057 to 2.678, relative to incomplete information.
The signal improves allocation, but it does not automatically improve user welfare.

03 Impact & dialogue

Potential Application: Transaction-Priority Interfaces

Wallets or transaction interfaces may help users choose fees without observing other users' urgency. Our model tests whether nonbinding differentiated recommendations can better align scarce confirmation priority with urgency.
Potential Benefit: Fewer same-fee collisions and better priority alignment.
Potential Cost: Higher fee expenditure and possible welfare loss.

Limitations & next test

Limitations: The model uses two users, three rounds, discrete fees, and stylized payoffs. Current results are simulations, not human-participant or live-network evidence.
Next Test: Collect participant choices and compare their beliefs, waiting responses, and signal compliance with the strategic and computational benchmarks.
If Behavior Differs: Revise the comparison policy or signal design and rerun the computational analysis.

Real-World Boundary

Algorand motivates the transaction-priority setting, but the simulation does not estimate current Algorand network behavior or justify deployment.

Acknowledgments

We thank Prof. Luyao Zhang and Prof. Kenneth S. Rogerson for their academic guidance, and Han Zhang for research-gap feedback. We thank Gihun Lee, Longfei Jing, Mustafa Ayub Khan, and Muhan Chen for their peer reviews and discussions. We also acknowledge Tencent's Shanghai office, the Shanghai Science and Technology Museum, and the COMSCI/ECON 206 classroom community for opportunities to explore technological innovation and interdisciplinary learning.

Selected references

Harsanyi, J. C. (1968). Games with Incomplete Information Played by Bayesian Players, Part II: Bayesian Equilibrium Points. Management Science, 14(5), 320–334.
Camerer, C. F. (2003). Behavioral Game Theory. Princeton University Press.
Gilad, Y., et al. (2017). Algorand: Scaling Byzantine Agreements for Cryptocurrencies. SOSP '17, 51–68.
Roughgarden, T. (2024). Transaction Fee Mechanism Design. Journal of the ACM, 71(4), Article 30.

OPEN MATERIALS

Hugging Face: https://huggingface.co/spaces/dku-comsci-econ206-2026/SubmitOrWait
GitHub: https://github.com/dku-comsci-econ206-Autumn2026/PS2-FP7-ZijieWang-LinZhang

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE